

M1 – Foundation of NLP

Single Selection

1.

Which of the following best describes the primary goal of lexical and syntactic processing in NLP?

- Converting audio to text efficiently
- Understanding the meaning and structure of text at the word and sentence levels
- Translating text from one language to another
- Storing textual data for retrieval

Single Selection

2.

Which of the following linguistic levels focuses on the structure and grammatical rules governing how words combine to form sentences?

- Morphology
- Syntax
- Semantics
- Pragmatics

Single Selection

3.

Traditional feature extraction methods such as Bag-of-Words and TF-IDF are limited because they:

- Are too computationally expensive
- Fail to capture the semantic meaning and context of words
- Do not require preprocessing like tokenisation
- Work only for speech data

Single Selection

4.

How do rule-based NLP systems differ from statistical or data-driven approaches?

- They are always more accurate.
- They learn from large labelled data sets.
- They depend on manually crafted linguistic rules and patterns.
- They use neural embeddings to represent text.

Single Selection

5.

Which of the following is an example of a real-world NLP application relying on both lexical and syntactic processing?

- CPU scheduling
- Image segmentation
- Sentiment analysis and conversational agents
- Database indexing

Lexical Processing

1.

What is the primary goal of text normalisation (e.g., lowercasing, punctuation handling, and whitespace cleanup) in lexical preprocessing?

- To create a consistent text representation that reduces superficial variation before analysis
- To guarantee that capitalisation-based meanings (e.g., 'US' vs 'us') are always preserved
- To permanently remove all numbers, emojis, and punctuation from any data set
- To replace domain knowledge so models do not need task-specific rules

Single Selection

2.

Which scenario best motivates the use of rules/regex-aware tokenisation rather than simple whitespace splitting?

- Processing a corpus where every word is separated by single spaces
- Splitting only on newline characters to obtain sentences
- Handling abbreviations such as 'Dr', multiword expressions such as 'New York' and contractions such as 'don't'
- Tokenising code comments that contain only lowercase words

Single Selection

3.

Which of the following is a sound strategy for managing stopwords across different domains?

- Always create a fixed, generic stopwords list for every task.
- Start with a base list and then customise for each task/domain; keep negations if the meaning depends on them.
- Retain all high-frequency words because frequency always signals importance.
- Remove interrogatives (such as 'what' and 'why') in question answering to reduce noise.

Single Selection

4.

Which of the following options correctly pairs the morphological process with its example?

- Derivation: 'run' → 'running' (same part-of-speech)
- Inflection: 'teach' → 'teacher' (verb to noun)
- Inflection: 'run' → 'runs' (grammatical change, core meaning preserved)
- Derivation: 'cats' → 'cat' (plural to singular)

5.

Which approach best preserves meaning while reducing word forms and handling irregular variants?

- Lemmatisation that uses a lexicon and words role in a sentence
- Porter/Snowball stemming that truncates suffixes via rule sets
- Edit distance that counts character edits between two strings
- Noisy channel scoring that ranks candidate corrections by prior and error likelihood

Syntactic Processing

Single Selection

1.

What is the primary objective of syntactic analysis in NLP?

- To identify how words combine into structures that express grammatical relations
- To translate sentences between languages word by word
- To convert audio signals into text transcripts
- To remove uninformative words from text

Single Selection

2.

Why is Part-of-Speech (POS) tagging important for downstream NLP tasks?

- It assigns grammatical categories to tokens, enabling parsing and information extraction.
- It guarantees accurate sentence meaning without context.
- It replaces the need for any grammar or parsing algorithms.
- It is only useful for languages without word boundaries.

3.

Which of the following statements best distinguishes common POS tagging approaches?

- Rule-based uses handcrafted rules, HMM learns tag sequences probabilistically and Brill learns error-fixing transformations.
- Rule-based predicts tags with neural embeddings, HMM relies on deterministic suffix rules and Brill is unsupervised clustering.
- Rule-based requires no lexicon, HMM ignores tag sequences and Brill applies random corrections.
- All three methods work identically but differ only in speed.

Single Selection

4.

What is a key difference between constituency and dependency parsing?

- Constituency groups words into phrase nodes, and dependency links words directly via head-dependent relations.
- Constituency uses only POS tags, and dependency uses only characters.
- Constituency cannot represent ambiguity, and dependency always produces multiple trees.
- Constituency is for speech only, and dependency is for written text only.

Single Selection

5.

Which of the following examples illustrates grammatical agreement checking in syntactic analysis?

- Detecting that 'These houses is old' should be 'These houses are old'
- Lowercasing 'The' to 'the' for consistency
- Removing the stopwords 'the' and 'is' to reduce noise

- Measuring edit distance between 'form' and 'from'

Graded Question

1.

Which of the following steps most directly converts cleaned text into a machine-usable numeric form for modelling in a classic NLP pipeline?

Single Choice

- Tokenisation
- Feature extraction (e.g., Bag-of-Words, TF-IDF and embeddings)
- Modelling
- Stopword removal

2.

Which of the following scenarios most strongly suggests using regex/rule-based tokenisation (or a robust library tokeniser) instead of simple whitespace splitting?

Single Choice

- Clean English text with single spaces and no abbreviations
- Text with 'Dr', '3.14', 'don't' and multiword expressions such as 'New York'
- A list of single, lowercase keywords separated by spaces
- Short labels that are already pre-separated and punctuation-free

3.

Suppose you are building a high-precision QA system. Which preprocessing choice is most appropriate for preserving meaning while unifying word forms?

Single Choice

- Use Porter/Snowball stemming for speed, accepting non-word stems
- Use dictionary-based lemmatisation with POS, accepting a higher compute cost
- Remove all suffixes manually with a fixed list, ignoring context
- Skip any reduction; rely only on raw surface forms

4.

Which of the following statements best contrasts rule-based, HMM and Brill POS taggers?

Single Choice

- Rule-based: Hand-written rules; HMM: Probabilistic sequence model; Brill: Learns ordered error-correcting transformations

- Rule-based: Neural embeddings; HMM: Character rules only; Brill: Random rewrites
- Rule-based: No lexicon; HMM: Ignores tag transitions; Brill: Unsupervised clustering
- All three are identical, and the differences are only in the runtime speed.

5.

Which pairing correctly matches the method with its primary representation or goal in syntactic analysis?

Single Choice

- Constituency parsing → Head-dependent arcs without phrase nodes
- Dependency parsing → Phrase-structure trees using CFG rule expansions
- PCFGs → Choose among multiple valid trees by assigning probabilities to grammar rules
- Chunking → Deep, fully nested trees that resolve long-distance dependencies

M2 -Semantic Processing and Classical Text Generation

Semantic Processing -Practice Set

1.

Which of the following best distinguishes between 'polysemy' and 'homonymy' in lexical semantics?

- Polysemy refers to multiple related meanings of a word, whereas homonymy refers to multiple unrelated meanings sharing the same form.
- Polysemy and homonymy are identical; both mean that a word has several unrelated meanings.
- Polysemy involves unrelated meanings, whereas homonymy deals with words from different languages.
- Polysemy and homonymy both describe words with the same spelling that share the same meaning but differ in usage

Single Selection

2.

In semantic role labelling, which word functions as the agent in the sentence 'The chef cooked the meal with care'?

- The meal

- The chef
- With care
- Cooked

3.

What principle does the distributional hypothesis in semantics express?

- Semantic similarity is measured by comparing spelling patterns.
- Every word has a fixed meaning independent of context.
- Words that occur in similar contexts tend to have similar meanings.
- Meaning is determined only by dictionary definitions.

4.

What key advantage does a Conditional Random Field (CRF) model provide for named entity recognition (NER) compared to independent classifiers?

- CRFs enforce valid label sequences by modelling dependencies between consecutive tags.
- CRFs ignore surrounding tokens, focusing only on single-word features.
- CRFs depend only on fixed gazetteer lists for identifying entities.
- CRFs produce random label transitions during training.

5.

How does WordNet support knowledge-based word sense disambiguation through semantic similarity?

- By training deep neural networks using large unlabelled corpora
- By comparing distances between synsets in its hierarchical structure to estimate sense closeness
- By using raw word frequency counts without considering their meaning
- By ranking senses randomly when glosses overlap

Text Representation and Unsupervised Modelling -Practice Set

Single Selection

1.

Which of the following is a key limitation of the Bag-of-Words (BoW) model in text representation?

- It ignores the order of words and the context between them.

- It automatically captures the semantic similarity between words.
- It reduces sparsity by learning dense representations.
- It can distinguish between 'dog bites man' and 'man bites dog'.

Single Selection

2.

In a Document–Term Matrix (DTM) generated from a Bag-of-Words representation, what does each cell typically contain?

- The weight of each document in a topic distribution
- The probability of a topic occurring in a document
- The semantic similarity between two terms
- The frequency or count of a specific term in a document

Single Selection

3.

Which statement best explains how TF-IDF, which stands for Term Frequency-Inverse Document Frequency, improves upon plain Bag-of-Words representation?

- TF-IDF measures syntactic similarity between documents.
- TF-IDF assigns higher weights to all words equally to avoid sparsity.
- TF-IDF assigns lower weights to very frequent words and highlights words unique to specific documents.
- TF-IDF replaces words with dense embeddings using neural models.

Single Selection

4.

What is the key motivation behind using word embeddings such as Word2Vec or GloVe instead of BoW or TF-IDF?

- To replace preprocessing steps such as tokenisation and stopword removal.
- To ensure every document has a unique vocabulary without overlap.
- To increase sparsity for faster matrix operations.
- To learn dense, low-dimensional representations that capture semantic and syntactic relationships between words.

Single Selection

5.

In Non-negative Matrix Factorisation (NMF), what is the main goal of the factorisation process?

- To approximate the original document–term matrix as the product of two smaller non-negative matrices representing document–topic and topic–term relations.
- To randomly assign topics to words and refine them through probability distributions.
- To compress text data by allowing both positive and negative word weights.
- To calculate topic probabilities using Dirichlet priors and Bayesian inference.

Use Cases of NLP -Practice Set

Single Selection

1.

Which of the following best describes a limitation of classical n-gram language models?

- They automatically learn semantic meaning through embeddings.
- They cannot capture long-range dependencies due to fixed-length context.
- They encode the sentences as dense, low-dimensional vectors.
- They eliminate the need for any smoothing techniques.

Single Selection

2.

What is the main goal of laplace smoothing technique in language modelling?

- To increase the vocabulary size of the corpus.
- To assign non-zero probabilities to unseen n-grams and improve generalisation.
- To remove low-frequency words from the corpus.
- To increase the number of training examples by duplication.

Single Selection

3.

What does a lower perplexity value indicate when evaluating a language model?

- The model uses fewer words in its vocabulary.
- The model has memorised the training data without generalising.
- The model predicts the test data more confidently and accurately.
- The model generates sentences with random probabilities.

Single Selection

4.

In classical Natural Language Processing, which statement best describes template-based text generation?

- It fills predefined sentence patterns with slot values using handcrafted rules.
- It generates text by sampling from probabilistic word embeddings.
- It relies on deep transformer networks for contextual coherence.
- It extracts sentences from multiple documents for summarisation.

Single Selection

5.

How did early statistical machine translation (SMT) systems improve upon rule-based translation methods?

- By learning bilingual phrase alignments and translation probabilities from large corpora
- By manually encoding grammar and dictionary rules for each language pair
- By relying on template substitution without statistical learning
- By using only monolingual text and ignoring parallel data

Graded Question

1.

What role does WordNet play in classical semantic processing?

Single Choice

- It is a statistical model that predicts word sequences in a sentence.
- It provides a structured lexical database of synonym sets and semantic relations for representing word meanings.
- It generates embeddings through neural networks for semantic similarity.
- It is used for grammatical parsing and dependency extraction only.

2.

How does the Lesk algorithm determine the correct sense of a word in a sentence?

Single Choice

- It relies on transition probabilities between part-of-speech tags.
- It computes cosine similarity between the embedding vectors of the words in a sentence.
- It identifies the dictionary sense whose gloss shares the most overlapping words with the surrounding context.
- It assigns senses based on the most frequent word usage in the corpus only.

3.

Why are Conditional Random Fields (CRFs) often preferred for sequence labelling tasks such as named entity recognition (NER)?

Single Choice

- Because CRFs model dependencies between adjacent tokens, they ensure valid and globally consistent label sequences.
- Because CRFs rely solely on classifying individual tokens without context.
- Because CRFs eliminate the need for linguistic features or training data.
- Because CRFs randomly assign labels to balance precision and recall.

4.

Which of the following correctly describes thematic or semantic roles in NLP?

Single Choice

- They are used to detect morphological roots and affixes in words.
- They are grammatical categories used only to determine part-of-speech tags.
- They assign probabilistic weights to words for topic clustering.

- They identify functions such as who performs an action (i.e., the agent) and who receives it (i.e., the patient) within a sentence.

5.

Which of the following best describes rule-based or template-based text generation in classical NLP systems?

Single Choice

- It generates sentences by filling predefined templates with variable values using handcrafted rules.
- It produces text by sampling next-word probabilities from large corpora.
- It learns sentence structure entirely from neural embeddings without rules.
- It summarises existing text by extracting frequent words and sentences.

Deep Learning

M1 -Foundation of Deep Learning -Practice Set

Understanding Neurons And Neural Network

Single Selection

1.

Which statement best captures the concept of 'synaptic plasticity' in biological learning?

- It allows neurons to change their firing rate randomly over time
- It fixes neural pathways permanently after training
- It creates new neurons each time learning occurs
- It adjusts the strength of connections based on neural activity

Single Selection

2.

In an artificial neuron, how do weights and bias work together to influence learning?

- Weights scale the inputs, and bias shifts the activation threshold
- Both weights and bias normalise the data before training
- Bias removes the need for an activation function
- Weights and bias determine the number of hidden layers

3.

How does a multi-layer perceptron (MLP) achieve hierarchical feature learning?

- By stacking layers that progressively learn from simple to complex patterns
- By repeating the same linear transformation across all layers
- By using a single hidden layer to memorise all the training data
- By increasing the neuron count without using activation functions

Single Selection

4.

Which of the following best illustrates how pattern recognition in neural networks parallels human cognitive learning?

- The network memorises every example, just as humans remember each past experience exactly
- The network generalises from past examples, similar to how humans recognise familiar faces in new situations
- The network ignores variation in input, similar to how humans focus only on identical patterns
- The network predicts outcomes purely through random trial without feedback

Single Selection

5.

Why are deep learning models particularly effective at complex pattern recognition tasks?

- They apply multiple non-linear transformations to extract layered features
- They rely on handcrafted features provided by experts
- They perform linear mappings between inputs and outputs
- They train without any optimisation or feedback mechanisms

Transitioning from Machine Learning to Neural Networks

-

Single Selection

1.

When does deep learning typically show a clear performance advantage over traditional ML approaches?

- When the data set is small and tabular
- When the task involves unstructured or high-dimensional data
- When feature engineering is carefully performed by experts
- When computation resources are extremely limited

Single Selection

2.

Which activation function is most suitable for a multiclass classification problem where the model must output probabilities that sum to 1?

- ReLU

- Tanh
- Softmax
- Sigmoid

Single Selection

3.

Which modification directly addresses the perceptron's failure to learn non-linear patterns?

- Adding more training epochs
- Introducing hidden layers with non-linear activation functions
- Removing the bias term
- Using a smaller learning rate

Single Selection

4.

In a neural network, what is the primary purpose of hidden layers?

- To combine simple features into higher-level abstractions
- To store weights permanently after training
- To reduce the size of the training data set
- To output the final class probabilities

Single Selection

5.

Why are tensors central to model development in PyTorch?

- They store only integer values efficiently

- They enable automatic differentiation and GPU computation
- They replace all NumPy arrays for text data
- They prevent model parameters from updating

Neural Network Implementation Essentials

Single Selection

1.

In a multi-layer perceptron (MLP), what happens during forward propagation?

- Gradients are computed and weights are updated
- Loss is backpropagated from output to input
- The model resets weights after each epoch
- Inputs pass through layers to generate predictions

Single Selection

2.

What is the main purpose of backward propagation when training a neural network?

- To calculate prediction accuracy after training
- To compute gradients and update weights based on loss
- To initialise all weights randomly before training
- To normalise data before passing it to the network

Single Selection

3.

Which loss function is most suitable for a multiclass classification problem?

- Mean Squared Error (MSE)
- Binary Cross-Entropy
- Categorical Cross-Entropy
- Mean Absolute Error (MAE)

Single Selection

4.

You are building a model to predict whether a tumor is benign or malignant (binary classification). Your model outputs probabilities, but you accidentally use Mean Squared Error (MSE) as your loss function instead of Binary Cross-Entropy. What will likely happen?

- The model will fail to train and throw an error
- The model will train but perform poorly compared to using Binary Cross-Entropy
- The model will perform exactly the same as Binary Cross-Entropy
- The model cannot output probabilities when using MSE

Single Selection

5.

Why is zero-weight initialisation discouraged in deep neural networks?

- It causes weights to grow too large, too quickly
- It reduces computational efficiency during training
- It prevents the network from converging to any minimum
- It leads to all neurons learning the same features

Model Generalisation Techniques

Single Selection

1.

How does L1 regularisation differ from L2 regularisation in its effect on model weights?

- L1 regularisation encourages sparsity by driving some weights to zero, whereas L2 regularisation penalises large weights smoothly
- L1 regularisation increases all weight magnitudes equally, whereas L2 reduces them randomly
- L1 and L2 regularisation both remove neurons from the network
- L1 regularisation only affects bias terms, whereas L2 affects weights

Single Selection

2.

Why should dropout be turned off during inference (testing)?

- It causes underfitting during evaluation
- It increases computational cost unnecessarily
- It introduces randomness, leading to inconsistent outputs
- It prevents gradients from being computed

3.

If a dropout rate is set too high (e.g., 0.8) during training, what is the most likely consequence?

- The network converges faster and achieves higher accuracy

- The network underfits because too many neurons are inactive
- The network memorises training data and overfits
- Dropout rate has no effect on learning

Single Selection

4.

Which of the following best explains the benefit of batch normalisation during training?

- It prevents any overfitting from occurring by freezing weights early
- It reduces the number of layers needed in a model
- It replaces activation functions such as ReLU or Sigmoid
- It standardises layer inputs, stabilising gradients and accelerating convergence

Single Selection

5.

What criterion is commonly used to decide when to apply early stopping?

- When training loss reaches zero
- When validation loss starts increasing for consecutive epochs
- When accuracy on training data improves continuously
- When all weights stop updating during backpropagation

Graded Question

1.

How does synaptic strengthening in biological neurons relate to learning in artificial neural networks?

Single Choice

- It is similar to increasing connection weights through training
- It corresponds to increasing the number of neurons in the network
- It represents the random activation of neurons to explore patterns in the data
- It equates to normalising inputs for each layer

2.

Which activation function would be most appropriate for the output layer of a binary classification model?

Single Choice

- ReLU
- Sigmoid
- Tanh
- Softmax

3.

In a deep neural network, what do earlier layers typically learn compared to later layers?

- Earlier layers learn simple features; later layers learn complex feature combinations
- Both learn identical high-level abstractions
- Earlier layers memorise training data; later layers generalise it
- Earlier layers handle optimisation; later layers perform data cleaning

4. During backward propagation, what occurs when gradients become very small in deep networks?

- The model trains faster
- The model experiences the vanishing gradient problem and learns slowly
- The model automatically adjusts its learning rate
- The model's weights explode uncontrollably

5.

What is a key benefit of applying batch normalisation in deep learning models?

- It eliminates the need for dropout complete

- It prevents the model from underfitting
- It ensures weights are initialised to zero
- It stabilises the learning process by reducing internal covariate shift

CNNs for Effective Image Analysis - Recall Quiz

Single Selection

1.

Why do traditional MLPs struggle with large image inputs?

- They cannot process numerical data efficiently.
- They treat every pixel as unrelated, losing spatial relationships.
- They automatically detect local patterns.
- They reduce the number of parameters with large images.

Single Selection

2.

What does spatial locality mean in images?

- Nearby pixels are usually related to each other.
- Every pixel is completely independent.
- Images are stored as random pixel values.
- The image must be flattened before use.

Single Selection

3.

Which feature of a CNN helps maintain spatial locality during image processing?

- Fully connected layers
- Convolutional filters
- Random weight initialisation
- Flattened layers

Single Selection

4.

How does the human visual cortex inspire CNN design?

- It uses recurrent signals for sequence learning.
- It processes all pixels equally in parallel.

- It builds complex perception from simple visual features.
- It relies on random connections between neurons.

Single Selection

5.

Which of the following best explains 'hierarchical feature learning' in CNNs?

- Features are learned from the output layer backwards.
- Simple features combine to form complex ones through multiple layers.
- Features are computed randomly for every image.
- Each layer repeats the same feature extraction process.

CNN Architecture Essentials - Recall Quiz

Single Selection

1.

What occurs during the convolution operation in a CNN?

- The filter scans small regions of the image to detect features.
- The filter replaces all pixel values with random numbers.
- The network reduces the image size without using filters.
- The filter converts the image into a single average value.

Single Selection

2.

What is the main purpose of a feature map in a CNN?

- To store the detected features from an input image
- To save the weights of all filters
- To reduce the total number of filters used
- To store the labels of training data

Single Selection

3.

What does a pooling layer do in a CNN?

- It reduces the size of the feature map but retains important information.
- It increases the number of filters in the model.
- It randomises pixel positions in the image.
- It removes all non-zero values from the feature map.

Single Selection

4.

Why do we flatten the output before sending it to dense layers?

- To convert multi-dimensional data into a one-dimensional (1D) vector for classification
- To remove unwanted features
- To reduce the number of neurons in the hidden layers
- To automatically detect more complex features

Single Selection

5.

What is the role of dense (i.e., fully connected) layers in a CNN?

- To combine extracted features and make final predictions
- To detect edges and corners in the image
- To downsample the feature map
- To remove noise from the image

Building & Training CNNs - Recall Quiz

Single Selection

1.

What is the main purpose of backpropagation when training a CNN?

- To initialise the weights of the convolutional layers
- To adjust weights based on the error between the predictions and actual values
- To convert the input images into a normalised form

- To freeze layers during transfer learning

Single Selection

2.

Why is data normalisation important when training CNNs?

- It speeds up learning and helps the model converge more easily.
- It increases the size of the training data_set.
- It reduces the number of convolutional filters required.
- It changes the labels to smaller numerical values.

Single Selection

3.

What is the goal of image augmentation during CNN training?

- To artificially increase data diversity and reduce overfitting
- To reduce image resolution for faster processing
- To remove unwanted images from the dataset
- To merge multiple images into one composite image

Single Selection

4.

What is transfer learning in the context of CNNs?

- Training a model from scratch on a new data_set
- Using a pre-trained model and fine-tuning it for a new task
- Moving data between different models
- Saving a trained model for later use

Single Selection

5.

What can visualising CNN activations help you understand?

- How each layer responds to different features in the an image
- The exact numerical value of every filter weight
- How the data_set was normalised before training
- Which hardware was used for training

Simple Applications of CNNs - Recall Quiz

Single Selection

1.

What does a CNN learn to do in an image classification task?

- Predict a single label that best describes the whole image
- Identify the exact location of objects in the image
- Divide the image into multiple segments based on colour
- Detect facial features and expressions

Single Selection

2.

How does object detection extend beyond image classification?

- It labels only one object per image.
- It identifies multiple objects and their positions.
- It removes background noise from the image.
- It groups similar images together.

Single Selection

3.

What is unique about image segmentation compared with object detection?

- It classifies the entire image into one label.
- It labels every pixel based on the object it belongs to.
- It detects only the largest object in the image.
- It counts the number of objects in the image.

Single Selection

4.

How do CNNs contribute to facial recognition systems?

- By learning unique patterns in facial features for identification
- By converting 3D images into 2D views
- By removing background objects from images
- By detecting moving faces in videos

Single Selection

5.

What problem does OCR solve using CNNs?

- Recognising text characters from images and converting them into digital form
- Detecting faces in low-light conditions
- Identifying objects and labelling them in a scene
- Matching different image styles together

RNNs for Sequences? - Recall Quiz

1.

What makes Recurrent Neural Networks (RNNs) different from traditional neural networks like MLPs?

- RNNs use multiple layers instead of a single one.
- RNNs process all inputs at once instead of step by step.
- RNNs can remember past information and use it to influence future outputs.
- RNNs ignore the order of inputs to simplify computation.

Single Selection

2.

Why do standard Multi-Layer Perceptrons (MLPs) struggle with sequential data like sentences or time series?

- MLPs lack a mechanism to remember past information.
- MLPs can only process numerical data.
- MLPs are slower than RNNs.
- MLPs require labelled data to train.

Single Selection

3.

What problem do basic RNNs face when handling long sequences?

- They become too slow to process.
- They tend to forget information from earlier steps in the sequence.
- They cannot be trained with backpropagation.
- They overfit easily to short sequences.

Single Selection

4.

Which of the following is **not** an example of an RNN application?

- Predicting the next word while typing a message.
- Detecting running or walking patterns using smartwatch motion data.
- **Classifying whether an image contains a cat or a dog.**
- Forecasting tomorrow's weather based on previous days.

Single Selection

5.

What key principle allows RNNs to understand patterns over time?

- Each new input is processed independently from others.
- **The network maintains a hidden state that carries information forward.**
- Inputs are shuffled to reduce overfitting.
- Earlier data points are permanently stored without update.

RNN Architecture - Recall Quiz

Single Selection

1.

What is the primary function of a Recurrent Neural Network (RNN)?

- Process images faster
- **Remember previous information in a sequence**
- Reduce the number of layers in a network
- Randomly shuffle data

Single Selection

2.

Which of the following is an example of a sequence?

- A single photo
- A random number
- A colour palette

- A sentence of words

Single Selection

3.

What does a single RNN cell do at each step in a sequence?

- Ignores previous steps and only looks at the current input
- Combines the current input with past memory to produce an output
- Deletes all previous data
- Creates multiple random outputs

Single Selection

4.

Why do RNNs use the same weights across all time steps?

- To make learning consistent across the sequence
- To increase the number of parameters
- To ignore the sequence order
- To process images

Single Selection

5.

Backpropagation Through Time (BPTT) in RNNs is used to:

- Shuffle the sequence randomly
- Process data without learning
- Make the network forget previous steps
- Train the network by updating weights based on errors over time

Limitations of Basic RNNs and Their Extensions - Recall Quiz

Single Selection

1.

What is a sequence in the context of RNNs?

- A single number

- A set of items in a specific order
- A random collection of data
- A network of neurons

Single Selection

2.

Why do basic RNNs face difficulties with long sequences?

- They always produce incorrect outputs
- They cannot process sequences at all
- Important details from earlier steps can fade away
- They only work with numbers, not words

Single Selection

3.

What is the main purpose of LSTM and GRU networks?

- Replace RNNs entirely
- Increase the number of layers in RNNs
- Help RNNs remember important information over longer sequences
- Reduce the size of datasets

Single Selection

4.

Which of the following is a key advantage of RNNs in sequence tasks?

- They ignore past information
- They convert sequences into random outputs
- They remember previous steps and use context to make predictions
- They can only process one input at a time

Single Selection

5.

In the context of RNNs, a sequence is:

- A random collection of data
- A single number

- A network of neurons
- A set of items in a specific order

Basic RNN Use Cases - Recall Quiz

Single Selection

1.

Which of the following is the key difference between sequence classification and sequence labelling?

- Sequence classification predicts a label for each element, whereas sequence labelling predicts a label for the whole sequence.
- Sequence classification predicts a single label for the entire sequence, whereas sequence labelling assigns labels to each element.
- Both tasks predict labels for individual elements, but labelling uses RNNs and classification doesn't.
- Sequence labelling is only used for time series data, while classification is only for text.

2.

In time series forecasting, why are RNNs particularly effective?

- Because they ignore past data to focus on the present.
- Because they can process data in parallel without considering order.
- Because they remember past events and understand the order of sequences.
- Because they can only model linear trends.

Single Selection

3.

Which of the following is an example of simple sequence labelling in NLP?

- Determining if a movie review is positive or negative.
- Predicting the next word in a sentence.
- Tagging words in a sentence as nouns, verbs, or adjectives.
- Forecasting monthly sales based on past data.

Single Selection

4.

What is the advantage of bidirectional RNNs in sequence labelling tasks?

- They can reduce the sequence length for faster processing.
- They consider both past and future context when labelling each element.
- They only focus on long-term dependencies and ignore short-term context.
- They can label entire sequences with a single output.

Single Selection

5.

Which of the following best describes an advanced aspect of sequence labelling?

- Ignoring the surrounding elements when predicting a label for one element.
- Using models to jointly estimate the likelihood of sequences of labels, such as with Conditional Random Fields.
- Only labelling the first and last elements in a sequence.
- Applying sequence labelling exclusively to non-textual data.

RNNs and NLP-Language as a Sequence - Recall Quiz

Single Selection

1.

What is the role of the hidden state in an RNN?

- It stores model parameters.
- It acts as evolving memory carrying context forward.
- It controls the learning rate.
- It directly predicts the next word in a sequence.

Single Selection

2.

Which of the following best explains why RNNs are suited for Natural Language Processing (NLP)?

- They process each input independently

- They ignore temporal order in data
- They maintain contextual information from previous inputs
- They require fixed-length inputs

Single Selection

3.

Which of the following best illustrates a short-range dependency in text?

- The relationship between “New” and “York” in “New York City”
- Connecting the subject and verb across a long paragraph
- Tracking tone across multiple sentences
- Identifying themes across a chapter

Single Selection

4.

What does Named Entity Recognition (NER) allow an RNN to do?

- Recognise and label names, places, and organisations in text
- Determine the overall emotion of a sentence
- Predict the next probable word
- Detect grammatical errors in writing

Single Selection

5.

Which RNN architecture is used for machine translation tasks?

- Sequence-to-sequence (encoder–decoder) model
- Feed-forward neural network
- Convolutional neural network
- Autoencoder

RNNs and NLP-Language as a Sequence - Graded Assessment

1.

Which of the following best explains why MLPs struggle with sequential data?

Single Choice

- MLPs can only process numerical data
- MLPs treat each input as independent, ignoring temporal relationships
- MLPs are designed to handle only categorical features
- MLPs have fewer parameters than RNNs

2.

What makes RNNs suitable for time-series forecasting?

Single Choice

- They store all input data in memory
- They apply convolution over time steps
- They maintain context from previous time steps through hidden states
- They use backpropagation only once per input

3.

What is meant by “vanishing gradient” in RNNs?

Single Choice

- Gradients become too large during training
- Gradients become too small to update weights effectively
- The model’s weights disappear after training
- The gradient only applies to output layers

4.

Which of the following best describes “language modelling”?

Single Choice

- Translating sentences between languages
- Predicting the next word in a sequence

- Identifying named entities in text
- Classifying text as positive or negative

5.

Which of the following techniques helps RNNs handle variable-length sequences?

Single Choice

- Regularisation
- Padding and masking
- Weight decay
- Batch normalisation